

# EMMANUEL BRYAN AUKO

## Full-Stack Enterprise Systems & Backend Engineer | Data Analytics Specialist

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### PROFESSIONAL SUMMARY

Accomplished **Full-Stack Systems & Backend Engineer** and **Data Analytics Specialist** with expertise in co-authoring Business Requirement Documents (BRDs), designing production-grade multi-tenant architectures, and deploying enterprise Ecosystem Banking platforms. Skilled in Java 21 / Spring Boot 4 microservices, Next.js 16 frontends, PostgreSQL schema optimization (52 Flyway migrations), Finacle Core Banking integrations, and banking-grade security (JWT, RBAC, Maker-Checker). Proven track record in operational data analytics, machine learning, and executive presentation.

### TECHNICAL EXPERIENCE

#### E&M; Tech Solutions (Emtech)

##### Full-Stack Systems & Backend Engineer

July 2026 – Present | Nairobi, Kenya

- **BRD Co-Authoring & Architecture:** Co-authored foundational Business Requirement Documents (BRDs) and designed hub-and-spoke data models for Ecosystem Banking, connecting Anchor Institutions with supplier/customer networks.
- **Multi-Tenant Backend Engineering:** Architected multi-tenant backend microservices using Java 21 & Spring Boot 4, executing 52 Flyway schema migrations, PostgreSQL triggers, and append-only tamper-proof audit logging.
- **Enterprise Security Governance:** Engineered banking-grade security architecture including JWT auth with Redis session backing, fine-grained Role-Based Access Control (RBAC), and dual-authorization *Maker-Checker* workflows.
- **Modern Next.js Frontend:** Built responsive Next.js 16 (App Router) / React 19 web interfaces featuring Zustand state management, Zod validation, and standardized component libraries (EcosystemPicker, CustomerPicker).
- **Core Banking Systems Integration:** Integrated Emtech Core APIs and Finacle Core Banking architecture via typed adapter layers, environment-based credential management, and OpenAPI/Swagger documentation.
- **DevOps & Automated Pipelines:** Configured containerized Docker deployment stacks, Nginx reverse proxies, and Jenkins CI/CD automation pipelines for continuous integration and delivery.

#### Intevia Ltd

##### Software Engineering Intern

Oct 2025 – Feb 2026 | Nairobi, Kenya

- Assisted in designing, developing, and maintaining enterprise backend services utilizing Python, SQL, and RESTful web services.
- Implemented automated data pipelines and ETL routines, ensuring high data accuracy, schema validation, and system reliability.
- Applied core algorithms and optimized data structures to enhance database query performance and data retrieval speeds.
- Authored Software Requirement Specifications (SRS), technical documentation, and executed unit and integration test suites.
- Leveraged advanced AI development tools (LangChain, Gemini) to accelerate workflow prototyping and developer productivity.

#### Podii Software Solutions

##### Intern, Backend Engineering

May 2022 – Sep 2022 | Kisumu, Kenya

- Developed and maintained RESTful API services for financial and e-commerce platforms backed by PostgreSQL databases.
- Implemented secure authentication and input validation logic, enhancing system reliability and application security.
- Assisted in production issue triage, system log analysis, Git version control, and active peer code reviews.

### LEADERSHIP & FELLOWSHIP EXPERIENCE

#### Power Learn Project (PLP) & KPC Foundation

##### Data Analytics & AI Fellow / Hackathon Winner

Oct 2025 – Present | Nairobi, Kenya

- Selected for an intensive 12-week fellowship focused on operational data analytics, machine learning, and predictive maintenance for midstream energy pipelines.
- **Hackathon Champion:** Won program hackathons and built an end-to-end operational analytics capstone solution calculating critical business KPIs and throughput metrics.
- Participated in executive KPC Masterclasses, mastering presentation skills, technical communication, and data storytelling for C-suite stakeholders.

#### Murang'a University Engineering Society

##### Association Lead / Mentor

Sep 2023 – Aug 2024 | Murang'a, Kenya

- Mentored peer engineers in modern Git collaboration, Spring Boot backend development, and relational database normalization.
- Defined engineering standards and architectural best practices across student-led software development projects.

EDUCATION & RESEARCH PROJECTS

**Murang’a University — Murang’a, Kenya**  
**BSc. in Software Engineering (Second Class Honors Upper Division)**  
• **Undergraduate Project:** *Efficiency Analysis of Distributed Data Processing* (Jan 2024 – Aug 2024) — Evaluated performance bottlenecks, benchmarking algorithms, and architectural trade-offs affecting system throughput and scalability.

Sep 2021 – Aug 2024

**Oriwo Boys High School — Homabay, Kenya**  
**Kenya Certificate of Secondary Education (KCSE) — Grade: B+**

Jan 2017 – Dec 2020

TECHNICAL SKILLS & CERTIFICATIONS

|                      |                                                                                                                                                               |
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| Languages & Core:    | Java (Spring Boot), Python, JavaScript, TypeScript, SQL                                                                                                       |
| Backend & Security:  | Spring Boot 4, REST APIs, Spring Security, JWT, Redis Sessions, RBAC, Maker-Checker Architecture, AOP                                                         |
| Frontend Stack:      | Next.js 16 (App Router), React 19, TypeScript, Zustand, Tailwind CSS v4, React Hook Form, Zod, Recharts                                                       |
| Databases & Banking: | PostgreSQL, Flyway Schema Migrations (V1–V52), Finacle Core Banking Architecture, Multi-Tenant SACCOs & ERPs                                                  |
| Data Analytics & AI: | Python (Pandas, NumPy, Scikit-Learn), ETL Pipelines, Machine Learning, Predictive Maintenance, SAS Programming                                                |
| DevOps & QA:         | Docker, Docker Compose, Nginx Reverse Proxy, Jenkins CI/CD, Git, Postman, Testcontainers Integration Testing                                                  |
| Certifications:      | <b>Microsoft Azure Certified:</b> AI-901 (Azure AI Fundamentals) & DP-900 (Azure Data Fundamentals)<br><b>SAS Certified:</b> SAS Programming & Administration |
| Affiliations:        | Associate Member — Kenya Computing and Information Technology Society (KCITS)                                                                                 |

REFEREES

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| <b>Dr. Victor Oriko</b><br>Software Engineering Lead<br>Intevia Limited & Microsoft<br>Email: info@intevia.co.ke<br>Mobile: 0721458657 | <b>Mr. Sigu Magwa</b><br>Team Lead Manager<br>Podii Consultants Limited<br>Email: sigu@podiihq.com<br>Mobile: 0733000003 | <b>Mr. Frank Midigo</b><br>Lead Backend Engineer<br>Podii Consultants Limited<br>Email: midigofrank@gmail.com<br>Mobile: 0741934955 |
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